

Ahmed Magd Aly Shehata

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RESEARCH INTERESTS

Theoretical reinforcement learning, optimization, robust sequential decision-making, constrained and utility-based RL, robot learning, embodied agents.

EDUCATION

Simon Fraser University

Sept 2025 – Present

Ph.D. in Computer Science

Advisor: [Sharan Vaswani](#). Research focus: theoretically principled reinforcement learning and optimization.

Korea Advanced Institute of Science and Technology (KAIST)

Aug 2021 – Aug 2023

M.S. in Robotics (AI and Computer Vision)

GPA: 4.05/4.30

Thesis: *Exploring Diffusion Models for Semantic Segmentation in Bird's Eye View Mapping for Autonomous Vehicle Perception*.

Innapolis University

Aug 2020 – Jun 2021

M.S. in Robotics and Computer Vision (transferred)

GPA: 4.67/5.00

Nile University

Aug 2016 – Aug 2020

B.S. in Mechatronics Engineering

GPA: 3.98/4.00 (2/110)

Thesis: *Intelligent EMG-Assisted Continuous Knee Motion*.

RESEARCH EXPERIENCE

Simon Fraser University

Jul 2025 – Present

Ph.D. Student, School of Computing Science

Advised by [Sharan Vaswani](#). Developing theoretically principled reinforcement learning methods, with current work centered on utility-based and constrained RL objectives, primal-dual optimization, and scalable algorithms for controlling agent behavior under non-standard objectives.

Johns Hopkins University

Apr 2024 – Aug 2024

Research Intern, Computer Science Department

Advised by [Zongwei Zhou](#) at the [CCVL Lab](#). Worked on medical image analysis and image synthesis using computer vision and generative models.

KAIST

Jun 2023 – Mar 2024

Researcher, Machine Learning & Mind Lab

Advised by [Sungjin Ahn](#).

- Worked on world models and model-based reinforcement learning.
- Improved long-horizon memory in sequence-model-based world models using RetNet-style architectures.
- Explored fine-tuning diffusion-model-based planning pipelines built on [Diffuser](#).

KAIST

Aug 2021 – Aug 2023

Graduate Student Researcher, VDC Lab

Advised by [Dongsuk Kum](#). Developed a diffusion-based model for semantic map prediction in bird's-eye view for autonomous driving perception.

Nile University

Jun 2019 – Jul 2020

Undergraduate Researcher

Advised by [Hossam Hassan Ammar](#). Conducted research at the intersection of robotics and machine learning, leading to multiple publications.

INDUSTRY EXPERIENCE

PLAIF

Oct 2024 – Jun 2025

Machine Learning & Robotics Engineer (Robot Learning)

- Develop reinforcement learning and imitation learning pipelines for real-world robotic manipulators.
- Work on vision, policy learning, and motion-control components to improve robustness and reliability for industrial deployment, including sim-to-real transfer and adaptation issues.

SELECTED RESEARCH PROJECTS

- **Utility-Based and Constrained Reinforcement Learning (SFU):** Developing scalable algorithms for reinforcement learning under non-standard utility and constraint-based objectives, with a focus on optimization-based methods that better control what behaviors are encouraged or avoided.
- **RetNet World Models (KAIST):** Modified sequence-model-based world models to improve long-horizon memory, targeting better planning and decision-making over long horizons in model-based RL.
- **Diffusion-Based Planning / RL (KAIST):** Investigated fine-tuning diffusion-model-based planning pipelines derived from Diffuser for reinforcement learning and sequential decision-making tasks.
- **Autonomous Driving Perception (KAIST M.S. thesis):** Built a diffusion-based model for semantic map generation in bird's-eye view for autonomous vehicle perception.
- **Robot Learning for Manipulation (PLAIF):** Develop learning-based control pipelines for dual-arm and industrial manipulation tasks using reinforcement and imitation learning, with emphasis on robustness for real-world deployment.

PUBLICATIONS

- A. Bangunharcana, **A. Magd**, K. S. Kim. *DualRefine: Self-Supervised Depth and Pose Estimation Through Iterative Epipolar Sampling and Refinement Toward Equilibrium*. CVPR 2023. [\[link\]](#)
- A. Sayed, A. A. Mohamed, **A. Magd**, et al. *Experimental Modeling of Hexapod Robot Using Artificial Intelligence*. AICV 2020. [\[link\]](#)
- H. Elkholi, A. T. Azar, **A. Magd**, et al. *Classifying Upper Limb Activities Using Deep Learning*. AICV 2020. [\[link\]](#)
- A. T. Azar, **A. M. Aly**, A. S. Sayed, et al. *Neuro-Fuzzy System for 3-DOF Parallel Robot Manipulator*. NILES 2019. [\[link\]](#)

HONORS & AWARDS

- Simon Fraser University funding and awards: Special Graduate Entrance Scholarship (\$10,000), Graduate Fellowship (\$7,000), CMPT Graduate Fellowship (\$2,000), and PhD Research Scholarship (\$3,672), 2025–2026.
- Full scholarship recipient, KAIST M.S. program (2021–2023).
- Full scholarship recipient, Innopolis University (2020–2021).
- Funding support for graduation project, Academy of Scientific Research and Technology, Egypt (\$10,000), 2020.
- Bank of Egypt full scholarship recipient, Nile University (2016–2020).
- Finalist, International Data Analysis Olympiad (Yandex IDAO), 2021. [\[link\]](#)
- 1st place, domain generalization deep learning contest, Innopolis University, 2020. [\[link\]](#)

SKILLS

Programming	Python, C/C++, Java, MATLAB
ML / Robotics	PyTorch, TensorFlow, OpenCV, ROS, reinforcement learning, imitation learning, constrained RL, diffusion models, computer vision
Systems	Linux/Unix, macOS, Windows, Git
Languages	Arabic (native), English (professional)